Ge 158 Lecture 8: Scattering from the Ocean and Other Surfaces with Small-Scale Roughness

Brent Minchew

Topics covered

- Ocean wave spectra
- Scattering from slightly rough surfaces (Bragg scattering)
- Scattering from undulating, slightly rough surfaces

1 Overview

- We will use scattering of EM waves from the ocean surface as an illustrative example of scattering from a broad range of natural surfaces.
- Advantages of doing so:
  - allows us to incorporate concepts from the beginning of the semester on dielectric properties (Lecture 1), because temperature and salinity have important effects on scattering from the ocean;
  - nicely illustrates the influence of surface roughness of varying wavelengths and sources.
- To get started, let us consider the geometry of the surface and the processes that influence it.

2 The ocean surface

2.1 What sets the elevation of the ocean surface

- The elevation of the ocean surface is influenced by a variety of factors. Roughly in order of decreasing wavelength:
  - the gravitational equipotential surface (geoid), which approximates mean sea level and is the elevation that the ocean would take given a planet's mass distribution and rotation, and in the absence of external forcing (tides, winds, etc.);
  - tides;
  - atmospheric pressure gradients;
  - currents;
  - ★ wind-driven waves → of greatest interest in today's lecture.

2.2 Wind-driven surface waves

- Ocean surface waves are mechanical waves that exist at the ocean–atmosphere boundary.
- As with any mechanical wave, they are defined by a perturbing force and a restoring force.
- In all cases of interest in this lecture, the perturbing forces are shear stresses caused by the wind.
- We will focus our attention on two types of waves defined by different restoring forces:
  1. Gravity waves are long wavelength (> 2 cm) whose restoring force is gravity.
     - Swells, which build up from wind forcing over long timescales, are examples of gravity waves.
     - Can play an important role in scattering from the ocean surface.
     - Terrestrial analogs: rolling hills, crop rows.
  2. Capillary waves are short wavelength (< 2 cm) whose restoring force is surface tension.
     - Produce small-scale roughness that is superimposed on longer-wavelength roughness.
     - Play a leading role in EM scatter from the ocean surface.
     - Terrestrial analog is any small-scale roughness: rocks, small erosional features.
- At radar wavelengths, it is often the case that the portion of the wave spectrum of greatest interest lies at intermediate wavelengths.
- At these intermediate wavelengths, both gravity and surface tension act to restore the ocean surface $\rightarrow$ such waves are called gravity–capillary waves.

2.3 Dispersion relation for gravity–capillary waves

- The dispersion relation for gravity–capillary waves in deep water (depth $\gg$ wavelength) is given as

$$\omega^2 = gK + \frac{\gamma_{\rm st}}{\rho_w} K^3, \tag{1}$$

  where

  - $g \equiv$ gravitational acceleration;
  - $\gamma_{\rm st} \equiv$ surface tension;
  - $\rho_w \equiv$ mass density of water;
  - $\omega \equiv$ angular frequency and $K \equiv$ wavenumber ($= 2\pi/\Lambda$) of the ocean-surface wave.
- In the limits
  - small K: $\omega \rightarrow \sqrt{gK}$ for gravity waves;
  - large K: $\omega \rightarrow \sqrt{\gamma_{\rm st} K^3/\rho_w}$ for capillary waves.
- We can see from this expression how "surfactants" (substances that reduce surface tension) change the dispersion relation of ocean surface waves $\Rightarrow$ change the scattering properties of EM waves.
- Common examples of surfactants include mineral oil (as in the oil spill example that we have seen many times so far in this course) and biological films. Microplastics may also act as a surfactant, but I believe this remains an open question.

2.4 Phase velocity

- Gravity–capillary waves are excited (grow) in a turbulent wind field and reach equilibrium when the wind speed matches the phase velocity of the surface waves. From the dispersion relation, we define the phase velocity

$$u_p = \frac{\omega}{K} = \sqrt{\frac{g}{K} + \frac{\gamma_{\rm st}}{\rho_w} K}. \tag{2}$$

- In general, surfactants will reduce the surface tension of clean water ⇒ (all else being equal) will slow (gravity–)capillary waves (Figure 1).

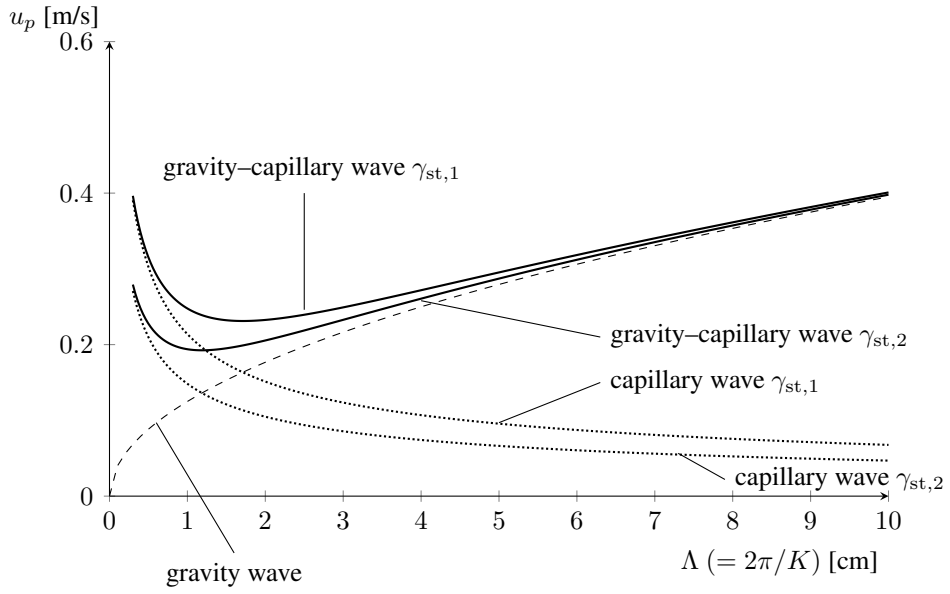


Figure 1: Phase velocity of deep-water gravity–capillary waves (solid, Eq. 2) versus wavelength for clean water ($\gamma_{\rm st,1} = 0.0728$ N/m) and for water whose surface tension has been reduced by a surfactant ($\gamma_{\rm st,2} = 0.035$ N/m, illustrative). Dashed: gravity-wave limit $\sqrt{g/K}$; dotted: capillary-wave limits $\sqrt{\gamma_{\rm st} K/\rho_w}$.

2.5 Wave height, the wave spectrum, and wind

- When wave speed (phase velocity) matches wind speed, the RMS height h of the ocean can be defined as

$$h^2 = \int_0^\infty \Psi(\omega)\, d\omega, \tag{3}$$

where the frequency spectrum Ψ is related to the local wind speed U (measured 19.5 m above the sea surface) through the empirical relation

$$\Psi(\omega) = \frac{8.1 \times 10^{-3}\, g^2}{\omega^5} \exp\left\{-\frac{3}{4}\left(\frac{g}{U\omega}\right)^4\right\} \tag{4}$$

⇒ strong correlation between wind speed and surface height, which, of course, we all know from experience. (Eq. 4 is the Pierson–Moskowitz spectrum for a fully developed sea.)

- For remote sensing, we are concerned with the direction of the winds relative to the look direction (or plane of incidence of the EM wave) of the sensor. This relation is usually given as an empirical relation between the normalized radar cross section σ^0_{pp} and the angle ϕ_{wind} between the plane of incidence and the wind direction:

$$\sigma^0_{pp} = AU^n \left(1 + a\cos\phi_{\text{wind}} + b\cos 2\phi_{\text{wind}}\right), \tag{5}$$

where A, a, b, n are all inferred from observations. This inference requires data collected from multiple look directions to provide info on wind speed and direction.

3 Simple models of EM scatter from the ocean (and other randomly rough surfaces)

- Let's dig a bit more into the mechanisms of EM scattering from the ocean surface.
- We'll start with the most useful model for surface scattering (at least at microwave frequencies): Bragg scattering or the small-perturbation model.
- Applicable when the RMS height of the surface and the horizontal length scale of surface roughness are small compared with the EM wavelength; canonically

$$kh < 0.3 \qquad \text{and} \qquad \frac{h}{\ell} < 0.3 \tag{6}$$

(Ulaby and Long (2014), p. 428, also require $k\ell < 3$).

- In general, h corresponds only to the component of the roughness spectrum responsible for the majority of scattering $\rightarrow$ called the Bragg wavelength.
- Basic idea of the model: any random surface can be decomposed into a series of sinusoids. The wavelength that contributes most to the backscatter power is the one that maximizes constructive interference (resonance) of scattered waves (Figure 2).

3.1 Bragg scattering

- From geometry, we find

$$\Lambda_{\text{Bragg}} = \frac{\lambda}{2\sin\theta_i} \tag{7}$$

or

$$K_{\text{Bragg}} = 2k\sin\theta_i = \frac{2\pi}{\Lambda_{\text{Bragg}}} \equiv \text{Bragg wavenumber.} \tag{8}$$

3.2 The first-order small-perturbation model

- To apply the concept of Bragg scattering to backscattered power, Rice (1951) applied perturbation methods and derived an expression for the first-order (meaning, higher-order terms in the Taylor series expansion are dropped) normalized radar cross section

$$\sigma^0_{pp} = 8k^4h^2\cos^4\theta_i \left|\alpha_{pp}\right|^2 W\left(K_{\text{Bragg}}, 0\right), \tag{9}$$

where $W(k_x, k_y) \equiv$ surface roughness spectrum discussed in the previous lecture

$$\left(W = \frac{\ell^2}{2}\exp\left\{-\left(\ell k\sin\theta_i\right)^2\right\} \quad \text{for a Gaussian surface}\right). \tag{10}$$

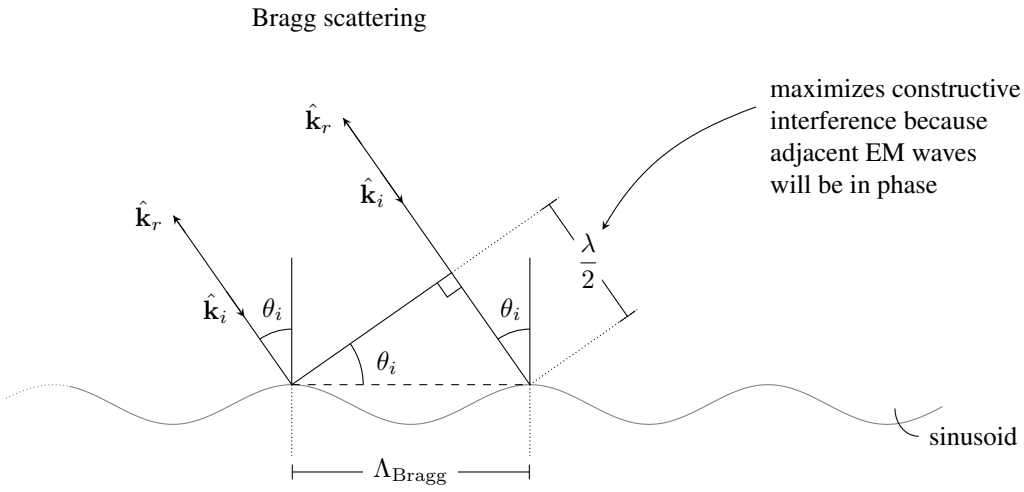


Figure 2: Geometry of Bragg scattering from a sinusoidal surface of wavelength Λ_{Bragg} for a wave incident at θ_i. The path to the farther crest is longer by $\Lambda_{\mathrm{Bragg}} \sin\theta_i = \lambda/2$ each way, so the backscattered waves from adjacent crests differ in path by one EM wavelength and add in phase.

- The scattering coefficients are

$$\alpha_{hh} = \rho_h \equiv \text{Fresnel reflection coefficient for } h\text{-polarized waves at oblique incidence,} \tag{11}$$

$$\begin{aligned} \alpha_{vv} &= \rho_v - \tau_v^2 \frac{(\varepsilon_2 - 1)\tan^2\theta_i}{2} \\ &= (\varepsilon_2 - 1)\frac{\sin^2\theta_i - \varepsilon_2\left(1 + \sin^2\theta_i\right)}{\left(\varepsilon_2\cos\theta_i + \sqrt{\varepsilon_2 - \sin^2\theta_i}\right)^2}, \end{aligned} \tag{12}$$

  where ρ_v and τ_v are the Fresnel reflection and transmission coefficients for v-pol at oblique incidence (Lecture 5), and we have assumed $\varepsilon_1 \approx 1$.

3.2.1 Why does $\alpha_{hh} = \rho_h$ but $\alpha_{vv} \neq \rho_v$?

A. Go back to EM boundary conditions: recall from Faraday's law we got the condition that the tangential component of the electric field is continuous across the surface (Lecture 4); the h field is entirely tangent to the surface for all roughness (at least to a reasonable approximation), whereas the tangential component of the v field changes with roughness.

3.2.2 Key take-aways

1. $\sigma^0_{hv} = \sigma^0_{vh} = 0$ to first order (higher-order factors introduce small nonzero values of σ^0_{hv} (and σ^0_{vh}), but these are generally negligible as they are in the noise of RS instruments).
2. α_{pp} represents total scattered power and is
   (a) polarization-dependent;
   (b) a function of dielectric properties and viewing geometry (indirectly EM wavelength);
   (c) independent of characteristics of surface roughness.

3. W influences the direction of scattering and is
   (a) dependent on characteristics of roughness;
   (b) dependent on EM wavelength and viewing geometry;
   (c) independent of dielectric properties;
   (d) independent of polarization.
4. $\sigma^0_{vv} > \sigma^0_{hh}$ for $\theta_i > 0$ (at normal incidence $\alpha_{vv} = \alpha_{hh}$) due to the expression for reflectivity (thus, due to EM BCs).

4 Tilted Bragg scatter model

- The previous model only applies to horizontal surfaces with small-scale roughness.
- We can combine the Bragg model with the Kirchhoff model for undulating surfaces (Lecture 7) by (again) treating the surface as a bunch of randomly rough surfaces (infinite extent) that are tilted due to longer-wavelength roughness features.
- On the ocean, this simply means that we represent the wind-driven gravity–capillary waves with a Bragg scatter model and rotate our local coordinate system to account for the passage of gravity waves (swells).
- This is a very accurate, simple model of scattering from the ocean surface for incidence angles ranging from 30°–65°. (Lower incidence angles produce primarily specular scatter, while angles $> 65°$ involve different mechanisms that we won't discuss in this course because they are esoteric.)

4.1 Normalized radar cross section of a tilted patch

- If we define the tilting of the rough surface patches in terms of two angles
  – $\psi \equiv$ in-plane (plane of incidence) tilt,
  – $\zeta \equiv$ anti-plane (plane of incidence) tilt,

  it can be shown (Valenzuela, 1978) that the normalized radar cross section is

$$\sigma^0_{pq} = 8k^4h^2\cos^4\theta'_i\,|S_{pq}|^2\,W, \tag{13}$$

  where

$$W = W\big(2k\sin(\theta_i+\psi),\ 2k\cos(\theta_i+\psi)\sin\zeta\big), \tag{14}$$

  and the polarization-dependent reflectivities, written as squared magnitudes of the rotated scattering coefficients S_{pq}, are

$$|S_{pp}|^2 = \left|\left(\frac{\sin(\theta_i+\psi)\cos\zeta}{\sin\theta'_i}\right)^2\alpha_{pp} + \left(\frac{\sin\zeta}{\sin\theta'_i}\right)^2\alpha_{qq}\right|^2 \qquad \text{for } p\neq q, \tag{15}$$

$$|S_{hv}|^2 = |S_{vh}|^2 = \left(\frac{\sin(\theta_i+\psi)\sin\zeta\cos\zeta}{\sin^2\theta'_i}\right)^2|\alpha_{vv}-\alpha_{hh}|^2, \tag{16}$$

  with

$$\theta'_i = \cos^{-1}\big(\cos(\theta_i+\psi)\cos\zeta\big) \equiv \text{incidence angle measured relative to the surface normal}, \tag{17}$$

  and α_{hh} and α_{vv} as defined in the Bragg model (Eqs. 11 and 12), evaluated at θ'_i.

4.2 Take-aways

- This model reduces to the previous (untilted) model when $\psi = \zeta = 0$.
- Take-aways:
  - Tilting modifies both reflectivity and the effective roughness spectrum.
  - But the dependencies of reflectivity and roughness spectrum remain the same as in the Bragg model and also include tilt angles.
  - Cross-polarized scattering (σ^0_{hv}) is nonzero when the anti-plane tilt angle $|\zeta| > 0$ (and $\theta_i + \psi \neq 0$)
    $\rightarrow$ because in this case, neither the h-polarized nor v-polarized electric fields are locally tangent to the surface $\Rightarrow$ conservation of the tangential component of the $\mathbf{E}$ field results in a change of polarization.
- And those are the basic concepts of EM scattering from surfaces.

References

Rice, S. O. (1951). Reflection of electromagnetic waves from slightly rough surfaces. *Communications on Pure and Applied Mathematics*, 4(2–3), 351–378. doi:10.1002/cpa.3160040206.

Valenzuela, G. R. (1978). Theories for the interaction of electromagnetic and oceanic waves—A review. *Boundary-Layer Meteorology*, 13(1–4), 61–85. doi:10.1007/BF00913863.

Variables

Units are SI. Values are given for universal constants; "typical" values are representative, not definitive.

Symbol	Description	Units	Value
Ocean surface waves			
g	gravitational acceleration	m/s^2	9.81
γ_{st}	surface tension	N/m	0.0728 (clean water, 20°C)
$\gamma_{\mathrm{st},1},\ \gamma_{\mathrm{st},2}$	surface tension of clean water and of surfactant-covered water (Figure 1)	N/m	0.0728; 0.035 (illustrative)
ρ_w	mass density of water	kg/m^3	≈ 1000 (seawater ≈ 1025)
ω	angular frequency of an ocean-surface wave	rad/s	–
K	wavenumber of an ocean-surface wave	rad/m	–
Λ	wavelength of an ocean-surface wave, $2\pi/K$	m	gravity: > 2 cm; capillary: < 2 cm
u_p	phase velocity, ω/K	m/s	minimum 0.23 (clean water)
h	RMS height of the ocean surface (Sec. 3: of the Bragg-scale roughness)	m	–
$\Psi(\omega)$	frequency spectrum of the surface height	m^2 s	Eq. (4)
U	wind speed (19.5 m above the surface in Eq. (4))	m/s	–
ϕ_{wind}	angle between the plane of incidence and the wind direction	rad	–
$A,\ a,\ b,\ n$	empirical coefficients in Eq. (5)	varies	inferred from observations
Bragg scattering			
k	EM angular wavenumber in the medium above the surface	rad/m	$2\pi/\lambda$
λ	EM wavelength	m	–
θ_i	angle of incidence	rad	–
$\hat{\mathbf{k}}_i,\ \hat{\mathbf{k}}_r$	propagation unit vectors of the incident and backscattered waves	dimensionless	–
Λ_{Bragg}	Bragg wavelength of the surface roughness	m	$\lambda/(2\sin\theta_i)$
K_{Bragg}	Bragg wavenumber	rad/m	$2k\sin\theta_i$
ℓ	correlation length	m	–
σ^0_{pq}	normalized radar cross section, receive/transmit polarizations p, q	dimensionless	$\sigma^0_{hv} = 0$ (first order, untilted)
$W(k_x, k_y)$	roughness spectrum (Lecture 7)	m^2	–
$k_x,\ k_y$	spectral wavenumbers of the roughness spectrum (Lecture 7)	rad/m	–
$p,\ q$	receive and transmit polarization labels (h or v)	–	–
$\alpha_{hh},\ \alpha_{vv}$	first-order scattering coefficients	dimensionless	$\alpha_{hh} = \rho_h$
$\rho_h,\ \rho_v,\ \tau_v$	Fresnel reflection coefficients (h, v) and transmission coefficient (v)	dimensionless	–

Symbol	Description	Units	Value
$\varepsilon_1,\ \varepsilon_2$	dielectric constants above and below the surface	dimensionless	$\varepsilon_1 \approx 1$
Tilted Bragg model			
ψ	in-plane tilt of a rough patch (in the plane of incidence)	rad	–
ζ	anti-plane tilt of a rough patch	rad	–
θ_i'	local angle of incidence (from the patch normal)	rad	$\cos^{-1}[\cos(\theta_i + \psi)\cos\zeta]$
S_{pq}	rotated scattering coefficient of a tilted patch ($\lvert S_{pq}\rvert^2$: polarization-dependent reflectivity)	dimensionless	$S_{pp} = \alpha_{pp}$ if $\psi = \zeta = 0$